

CS-600: Software Design and Development

2-2 Activity: Integrating Spring JPA Into System Design

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Database Integration Using Spring JPA – Reflection

As part of this assignment, I practiced integrating a Spring Boot application with a PostgreSQL database with the help of Spring Data JPA. This concept is extremely valuable in today's world of software development since connecting an application with a database allows saving data, retrieving it, and managing other functions related to storing information. Prior to completing this task, I had some understanding about databases, but now I have seen how everything interacts and what roles certain layers play in making a functional system.

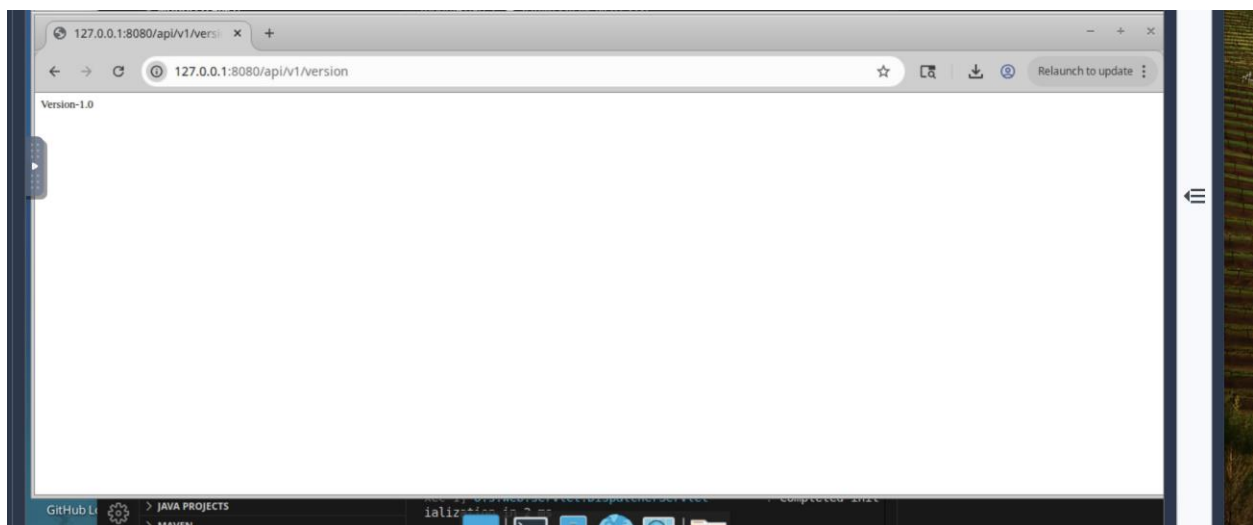
It should be emphasized that being able to integrate a database allows designing an application in a much better way. First, without using databases, all information will be saved during the operation of the application but will disappear as soon as it is closed. With PostgreSQL, the application can save data forever, and it becomes possible to retrieve it. The use of a database is crucial in virtually all applications, including those that must work with the data provided by users. In the current example, it is possible to retrieve names from the database using API endpoints.

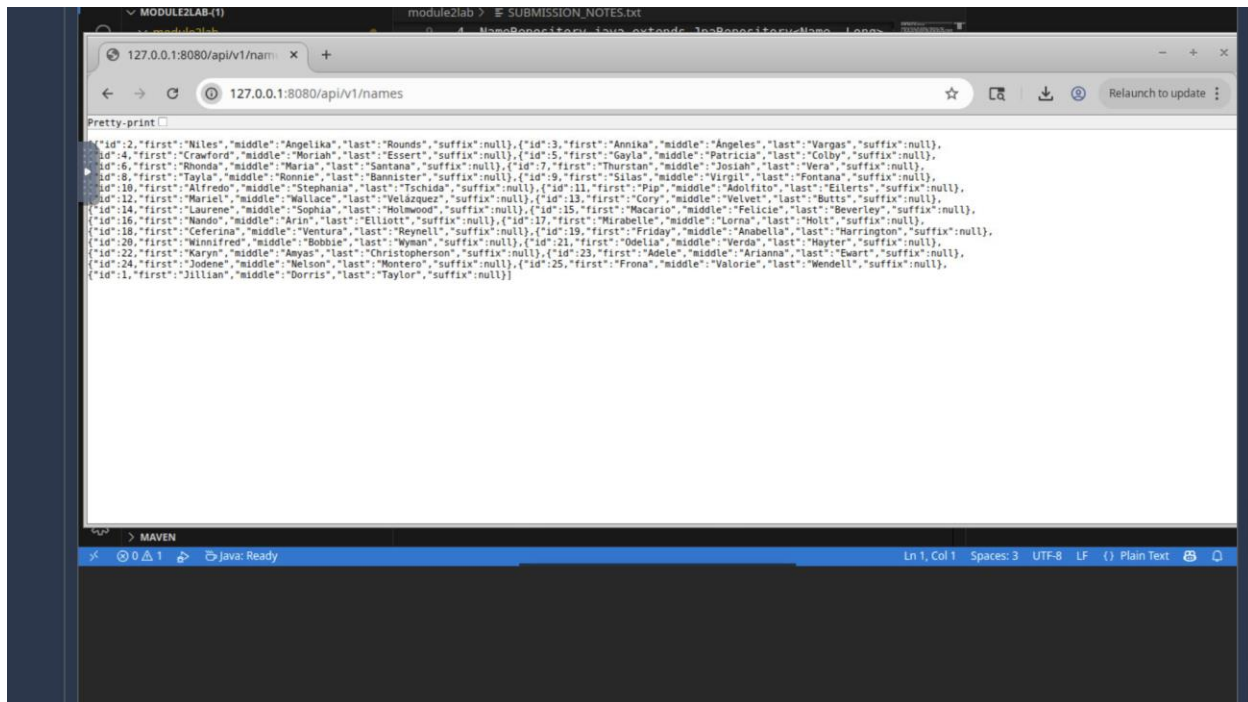
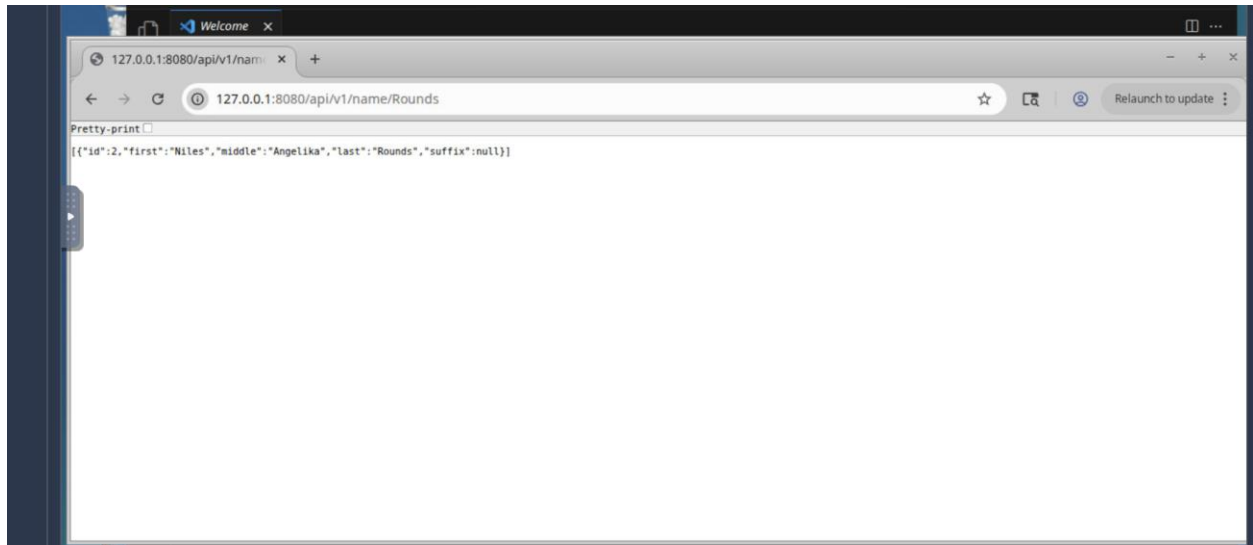
Another key aspect illustrated by this exercise is the importance of using a layered approach in developing backend systems. The current example consists of several layers that have different responsibilities: entity, repository, service, and controller. The entity layer defines the data model of the database. Then, a repository interacts with the database, executing various queries there. At the same time, the service layer provides a link between the repository and the controller, and

it can be used to apply business rules. Finally, the controller layer uses API endpoints to access the application.

The use of Spring Data JPA makes working with a database much simpler, and a developer must write fewer lines of code since it allows creating entities that correspond to tables in the database. In addition, it allows generating CRUD operations without having to write all the logic manually. In the current assignment, I used both the capabilities offered out-of-the-box and custom queries to extract data from the database. Thus, the assignment proved how convenient and powerful this tool really is.

Checking whether database integration worked or not was a major step, too. I ran the application and called the endpoint through the browser to make sure that the connection works. As a result, I received a list of records returned by the database in JSON format. Verifying the correctness of integration with a database is extremely important since this step allows for checking the proper functioning of all parts of the system.





This assignment is quite practical as it reflects the real aspects of working in software engineering. Modern applications usually rely on databases and learning how to integrate them is crucial. Therefore, this experience helps gain skills in using the tools that are typically used in

software development, including Spring Boot and PostgreSQL. At the same time, the assignment allowed learning more about how backend applications function.

I find this assignment personally significant due to my interest in developing impactful applications. These systems typically need to store data reliably and learning how to do this helps ensure that such systems can perform their functions well. For instance, an application designed to help victims will require saving valuable information, and database integration makes this possible. To conclude, I find that this assignment gave me valuable skills in using database integration, designing applications, and developing backend systems. It allowed us to see how layers interact and what kind of tools can be used.